

Column Space and Row Space

Linear Algebra - 24DS

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QUEST - 24DS

Motivation: Why these spaces?

- The **column space** tells us which vectors can be written as $A\mathbf{x}$ (Column space is simply output of a linear transformation).
- The **row space** is closely related to the null space (orthogonal complement).
- **Rank** (redefined) = dimension of column space = dimension of row space.
- Applications in:
 - Solving linear systems ($A\mathbf{x} = \mathbf{b}$ is solvable iff $\mathbf{b}$ is in the column space)
 - Data compression (low-rank approximations)
 - Principal Component Analysis (PCA)

Definitions

Definition (Column Space)

Let A be an $m \times n$ matrix. The **column space** of A , denoted $\text{Col}(A)$, is the span of the columns of A :

$$\text{Col}(A) = \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\} \subseteq \mathbb{R}^m$$

where $\mathbf{a}_j$ are the columns of A .

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Definition (Row Space)

The **row space** of A , denoted $\text{Row}(A)$, is the span of the rows of A :

$$\text{Row}(A) = \text{span}\{\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_m\} \subseteq \mathbb{R}^n$$

where $\mathbf{r}_i$ are the rows (as row vectors).

Key Facts

Important Theorems

- 1 $\dim(\text{Col}(A)) = \dim(\text{Row}(A)) = \text{rank}(A)$
- 2 The **pivot columns** of A (in RREF) form a basis for $\text{Col}(A)$.
- 3 The **nonzero rows** of $\text{RREF}(A)$ form a basis for $\text{Row}(A)$.
- 4 $\text{Row}(A) = \text{Row}(\text{RREF}(A))$ (row operations preserve row space).
- 5 Column operations change column space, but pivot columns method works directly on A .

How to Find a Basis for Column Space

Procedure:

- 1 Compute the reduced row echelon form (RREF) of A .
- 2 Identify the pivot columns in the RREF (columns containing leading 1's).
- 3 The **corresponding original columns** of A form a basis for $\text{Col}(A)$.

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Why original columns?

Row operations preserve linear relations among columns, but change the actual column vectors. The pivot columns of the original matrix are linearly independent and span the column space.

Example in $\mathbb{R}^2$: Column Space

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- Columns: $\mathbf{c}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\mathbf{c}_2 = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$
- Notice $\mathbf{c}_2 = 2\mathbf{c}_1$.
- RREF: $\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} \rightarrow$ pivot column 1 only.
- Basis for $\text{Col}(A)$: $\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}$.
- $\dim(\text{Col}(A)) = 1$ (a line through origin).

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- RREF: $\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} \rightarrow$ nonzero rows: $(1, 2)$.
- Basis for $\text{Row}(A)$: $\{(1, 2)\}$.
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Rank = 1. Column space and row space both have dimension 1.

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$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\text{Further simplify: } R_1 \leftarrow R_1 - 3R_2 \rightarrow \begin{pmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

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Further simplify: $R_1 \leftarrow R_1 - 3R_2 \rightarrow \begin{pmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. **Pivot columns:**

Columns 1 and 3.

Example in $\mathbb{R}^3$: Basis for $\text{Col}(B)$

Original matrix B :

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Pivot columns in RREF: column 1 and column 3.

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Basis for $\text{Col}(B)$

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \\ 10 \end{pmatrix} \right\}$$

Check: Are they independent? Yes (one is not a scalar multiple of the other). Column 2 = $2 \cdot \text{col}1$, Column 4 = $-2 \cdot \text{col}1 + 1 \cdot \text{col}3$.

Example in $\mathbb{R}^3$: Basis for Row Space of B

Matrix B again:

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Nonzero rows of RREF:

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Note: Row space is in $\mathbb{R}^4$ (since B has 4 columns). Dimension = 2.

Alternative Method for Column Space

Sometimes you can find basis directly by inspection:

- Look for linearly independent columns.
- For B above: col1 and col3 are clearly independent.
- Check $\text{col2} = 2 \cdot \text{col1} \rightarrow$ dependent.
- $\text{col4} = ?$ Solve $c_1 \text{col1} + c_3 \text{col3} = \text{col4}$.
- From RREF, we saw $\text{col4} = -2\text{col1} + \text{col3}$.

Caution

For complicated matrices, always use the RREF pivot method to be safe!

Another $\mathbb{R}^2$ Example: Full Rank

$$\text{Let } C = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Another $\mathbb{R}^2$ Example: Full Rank

Let $C = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

- Columns: $(1, 3)$ and $(2, 4) \rightarrow$ linearly independent.
- RREF: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow$ pivot columns 1 and 2.
- Basis for $\text{Col}(C)$: $\left\{ \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix} \right\}$. Dimension = 2.
- Row space basis from RREF nonzero rows: $\{(1, 0), (0, 1)\}$ or equivalently original rows $(1, 2)$ and $(3, 4)$.
- Rank = 2.

Summary: How to Find Bases

Space	Basis from
Column space	Pivot columns of original matrix
Row space	Nonzero rows of RREF

Steps:

- 1 Compute RREF of A .
- 2 Identify pivot columns $\rightarrow$ take those columns from **original** A for column space.
- 3 Take nonzero rows of RREF for row space.

Always: $\dim(\text{Col}(A)) = \dim(\text{Row}(A)) = \text{rank}(A)$.

Practice Exercise

Find bases for the column space and row space of

$$D = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}$$

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Hint: RREF is $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$.

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Hint: RREF is $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$.

Answer

- Column space basis: columns 1 and 2 of D : $\{(1, 1, 0), (1, 0, 1)\}$
- Row space basis: $\{(1, 0, 1), (0, 1, -1)\}$
- Rank = 2

Key Takeaway for CS

- In machine learning, the column space of the data matrix represents all possible linear combinations of features.
- Low-rank approximations (truncated SVD) keep only the most important columns.
- Row space tells us constraints that data must satisfy.
- Understanding these spaces helps debug why $A\mathbf{x} = \mathbf{b}$ might have no solution ($\mathbf{b}$ not in column space).

Always remember: The column space is where A sends vectors.